$x = x_i$ is given by $\pi y_i^2 \Delta x$, $i = 0, 1, 2, ..., n-1$. Summing all these elementary volumes, we get the approximate volume of the solid of revolution as $\sum_{i=0}^{n-1} \pi y_i^2 \Delta x$. Let n become larger and larger $(n \to \infty)$ such that Δx becomes smaller and smaller $(\Delta x \to 0)$. Then $\sum_{i=0}^{n-1} \pi y_i^2 \Delta x$ tends to the volume of the solid of revolution. Hence the volume of the solid of revolution is $\pi \int_a^b y^2 dx$.

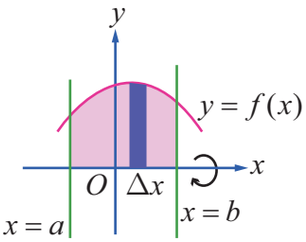


Fig. 9.36

Similarly, we can find the formula for finding the volume of the solid of revolution of the region bounded by the curve $x = f(y)$, y-axis, and the lines $y = c$ and $y = d$ about y-axis. The curve $x = f(y)$ lies to the right of y-axis between the lines $y = c$ and $y = d > c$. Assume that every line parallel to x-axis between $y = c$ and $y = d > c$ cuts the curve $x = f(y)$ in the first quadrant exactly at one point. See Fig.9.37**.** Then, the volume of the solid of revolution is given by $\pi \int_c^d x^2 dy$.

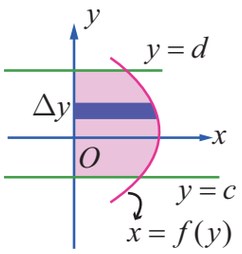


Fig. 9.37

Example 9.62

Find the volume of a sphere of radius a.

Solution

By revolving the upper semicircular region enclosed between the circle $x^2 + y^2 = a^2$ and the x-axis, we get a sphere of radius a. See Fig. 9.38**.**

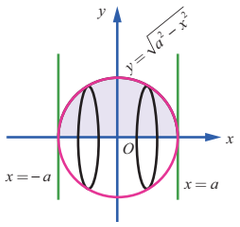


Fig. 9.38

The boundaries of the region are $y = \sqrt{a^2 - x^2}$, x-axis, the lines $x = -a$ and $x = a$. Hence, the volume of the sphere is given by

$$V = \pi \int_{-a}^{a} y^2 dx = \pi \int_{-a}^{a} (a^2 - x^2) dx$$

$$= 2\pi \int_0^a (a^2 - x^2) dx \text{, since the integrand } (a^2 - x^2) \text{ is an even function.}$$

$$= 2\pi \left(a^2 x - \frac{x^3}{3} \right)_0^a = 2\pi \left(a^3 - \frac{a^3}{3} \right) = \frac{4}{3} \pi a^3 .$$

Example 9.63

Find the volume of a right-circular cone of base radius r and height h.

Solution

Consider the triangular region in the first quadrant which is bounded by the line $y = \frac{r}{h} x$, x-axis, the lines $x = 0$ and $x = h$. See Fig.9.39**.** By revolving the region about the x-axis, we get a cone of base radius r and height h.

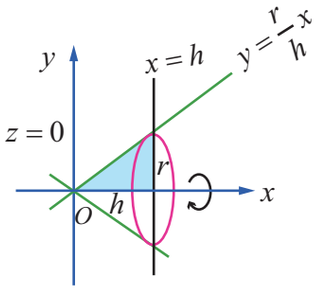


Fig. 9.39

Hence, the volume of the cone is given by

$$V = \pi \int_0^h y^2 dx = \pi \int_0^h \left(\frac{r}{h} x \right)^2 dx = \pi \left(\frac{r}{h} \right)^2 \int_0^h x^2 dx = \pi \left(\frac{r}{h} \right)^2 \left[\frac{x^3}{3} \right]_0^h = \frac{\pi r^2 h}{3}$$

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